

Inputs (public data)

Google Cloud WUE
(per region, cooling class)

EPA eGRID / ENE
(sub-regional Wlf)

WRI Aqueduct 4.0
(baseline water stress)

IEA stock + datasheets
(embodied water)

Model

Algorithm 1
$$W(f, r) = \sum \alpha_p * \beta_{p,f'} * V_{p,f',r} * S_r$$

Algorithm 2
Monte Carlo over S_r and WUE

Outputs

Scalar $W(f, r)$
+ 6-element attribution

Regional ranking
(vol vs stress vs carbon)

5/50/95 pct bands
(sensitivity)

Carbon-water
Pareto frontier